

CapitalMarket AI

AI-Powered Settlement Failure
Investigation Platform

Agentic Web

Orchestrate Actions

Team Introduction

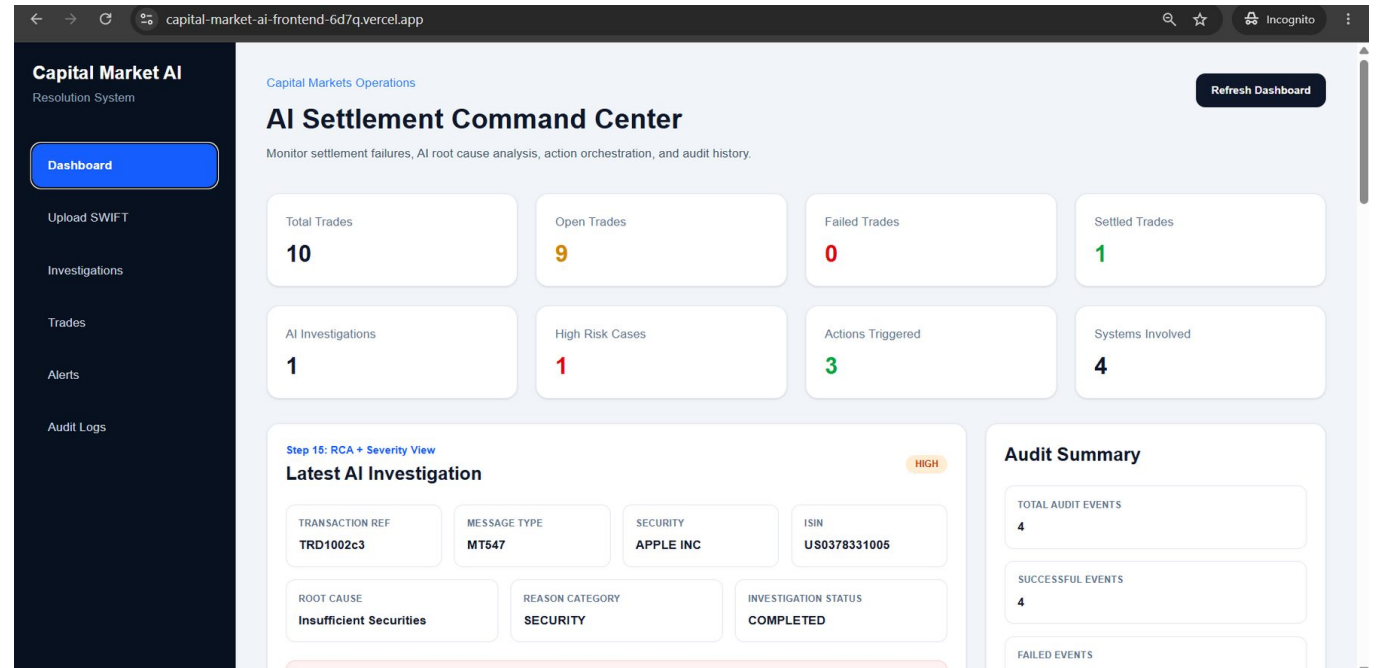
Anubhav Bangari

Full Stack Engineer | Capital Markets | AI Builder

Microsoft Build AI Hackathon 2026

Category: Orchestrate Actions Across Services

Live prototype dashboard



Problem Statement

Settlement failure investigation is manual, slow, and spread across systems.

Current Pain

Ops analysts manually check SWIFT confirmations, SSI instructions, cash balances, security holdings, and trade records.

Business Risk

Manual RCA creates delays, operational exposure, reconciliation effort, and missed settlement deadlines.

Hackathon Goal

Use an agentic workflow to investigate and trigger next actions.

Operational trade records

capital-market-ai-frontend-6d7q.vercel.app/trades

Incognito

Capital Market AI
Resolution System

Dashboard

Upload SWIFT

Investigations

Trades

Alerts

Audit Logs

Ops User
Operations

Trades

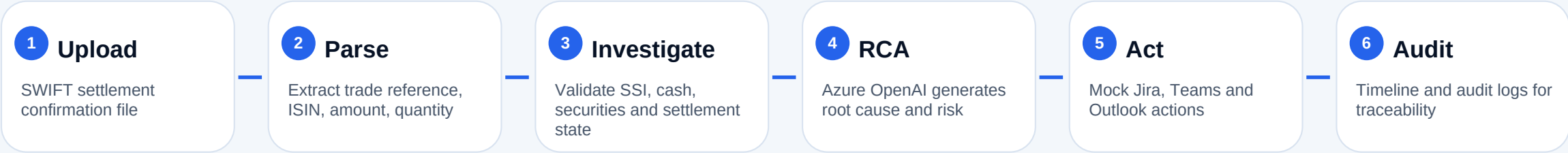
View settlement trade records.

Refresh

Trade Ref	Security	ISIN	Direction	Quantity	Amount	Status
TRD1001	MICROSOFT CORP	US5949181045	RECEIVE	1000.0000	USD 420000.00	OPEN
TRD1002	APPLE INC	US0378331005	DELIVER	5000.0000	USD 950000.00	OPEN
TRD1003	ALPHABET INC	US02079K3059	RECEIVE	3000.0000	USD 1200000.00	OPEN
TRD1004	AMAZON COM INC	US0231351067	DELIVER	1500.0000	USD 280000.00	OPEN
TRD1005	TESLA INC	US88160R1014	DELIVER	800.0000	USD 144000.00	OPEN
TRD1006	COCA COLA CO	US1912161007	RECEIVE	2000.0000	USD 125000.00	SETTLED
TRD1007	JPMORGAN CHASE	US46625H1005	DELIVER	4000.0000	USD 820000.00	OPEN
TRD1008	TCS LTD	INE467B01029	DELIVER	1000.0000	INR 4200000.00	OPEN
TRD1009	INFOSYS LTD	INE009A01021	RECEIVE	2500.0000	INR 3750000.00	OPEN
TRD1010	META PLATFORMS	US30303M1027	RECEIVE	1200.0000	USD 600000.00	OPEN

Solution Overview

A cross-platform workflow AI for settlement failure resolution.



Successful end-to-end upload

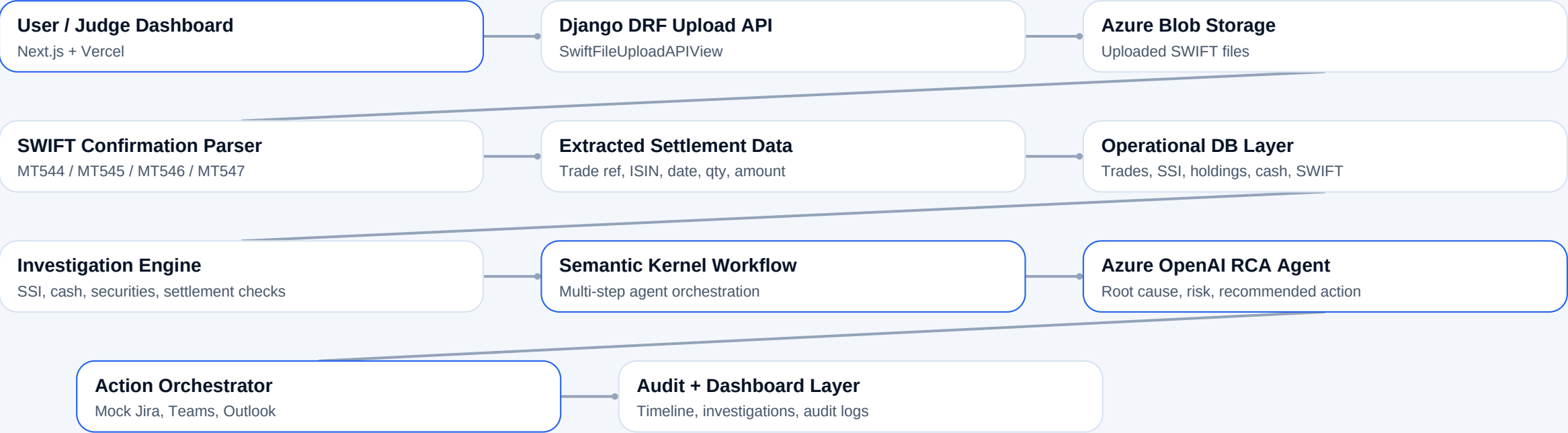
Parsed SWIFT Summary		
Message Type MT547	Transaction Ref TRD1002c3	SWIFT Message ID 1
Related Ref TRD1002	ISIN US0378331005	Security APPLE INC
Settlement Status FAIL	Quantity 5000	Amount USD 950000

Actions triggered by AI

Action Orchestrator		
AI triggered 3 operational actions across enterprise systems.		
JIRA P2 Settlement Failure - TRD1002c3 JIRA-TRD1002c3-SECURITY	TEAMS Settlement Alert - TRD1002c3 TEAMS-TRD1002c3	EMAIL Escalation Email - TRD1002c3 EMAIL-TRD1002c3

Architecture Diagram

Upload -> Parse -> Store -> Investigate -> AI RCA -> Action -> Audit



Microsoft stack proof: Azure OpenAI + Semantic Kernel + Azure App Service / Blob / SQL-ready architecture

AI Integration Details

Azure OpenAI and Semantic Kernel power the agentic workflow.

Semantic Kernel Workflow

Coordinates parser output, investigation checks, RCA generation, action orchestration, and audit updates.

Azure OpenAI RCA Agent

Generates root cause, confidence score, risk impact, and recommended operational action.

Failure Recovery

For SSI mismatch, cash shortage, securities shortage, or date mismatch, the agent suggests correction or escalation.

Azure OpenAI RCA Result

Capital Market AI

Resolution System

Dashboard

Upload SWIFT

Investigations

Trades

Alerts

Audit Logs

Capital Markets Operations

AI Investigations

View Azure OpenAI RCA, confidence score, risk impact, and multi-agent investigation trace.

Azure OpenAI RCA Result

HIGH

TRANSACTION REF

TRD1002c3

ROOT CAUSE

Insufficient Securities

CATEGORY

SECURITY

AZURE OPENAI

USED

MESSAGE TYPE

MT547

CONFIDENCE SCORE

90%

INVESTIGATION STATUS

COMPLETED

AI RCA Summary

The settlement investigation identified Insufficient Securities as the primary issue for transaction TRD1002c3. The case is categorized under SECURITY with HIGH severity.

Risk Impact

Potential settlement delay, operational follow-up, reconciliation effort, and market exposure.

Recommended Action

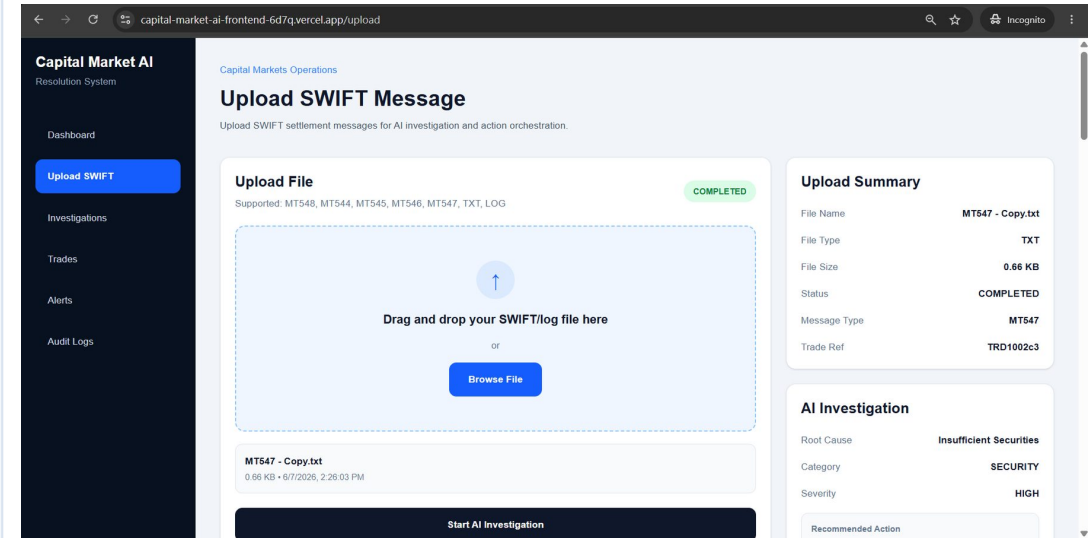
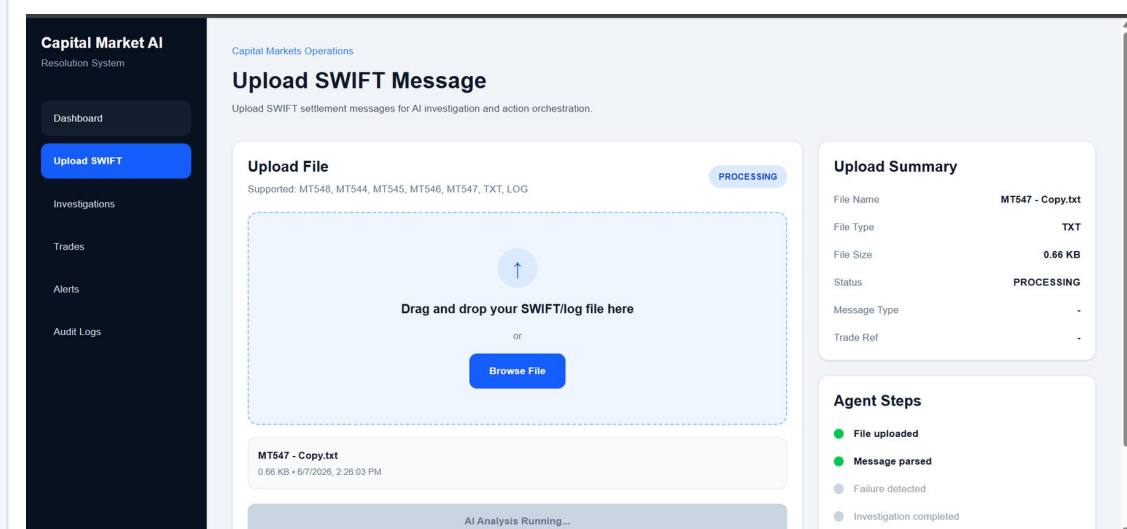
Acquire additional securities before settlement deadline.

Demo Screenshots: Upload & Processing

Judges can upload a SWIFT confirmation and watch the workflow progress.

Upload processing

Upload completed + investigation ready



Demo Screenshots: Parsing & Extracted Data

MT547 message parsed into settlement confirmation fields.

Parsed SWIFT summary

Upload Completed

File ID

1

Filename

MT547 - Copy.txt

Status

parsed

Duplicate

NO

File uploaded successfully

SWIFT file uploaded, parsed, investigated, and orchestrated successfully.

AI Investigation Result

Root Cause

Insufficient Securities

Category

SECURITY

Severity

HIGH

Recommended Action

Acquire additional securities before settlement deadline.

Upload completed

Parsed SWIFT Summary

Message Type

MT547

Transaction Ref

TRD1002c3

SWIFT Message ID

1

Related Ref

TRD1002

ISIN

US0378331005

Security

APPLE INC

Settlement Status

FAIL

Quantity

5000

Amount

USD 950000

Extracted Data Used by Agent

Message Type: MT547

Transaction Ref: TRD1002c3

Related Ref: TRD1002

ISIN: US0378331005

Security: APPLE INC

Settlement Status: FAIL

Quantity: 5000

Amount: USD 950000

Demo Screenshots: RCA & Action Orchestration

The agent converts investigation results into Jira, Teams, and email actions.

Enterprise escalation preview

Step 16: Mock Email / Teams Preview UI

Enterprise Escalation Preview

This shows exactly what the AI agent sent to Jira, Teams, and Email.

MOCK JIRA TICKET

JIRA-TRD1002c3-SECURITY COMPLETED

Issue Type: **Settlement Failure**

Priority: **P2**

Severity: **HIGH**

Root Cause: **Insufficient Securities**

Assigned To: **Settlement Operations Team**

Description

Trade TRD1002c3 failed because of Insufficient Securities. AI recommended action: Acquire additional securities before settlement deadline.

TEAMS ALERT PREVIEW

Settlement Alert COMPLETED

Settlement failure detected

Trade TRD1002c3 failed due to **Insufficient Securities**.

Severity: **HIGH**

Message Type: **MT647**

Category: **SECURITY**

AI Recommended Action

Acquire additional securities before settlement deadline.

Channel: #settlement-operations

EMAIL ESCALATION PREVIEW

Ops Escalation Email COMPLETED

To: ops@bank.com

CC: treasury@bank.com

Subject: **Settlement Escalation - TRD1002c3**

Hello Operations Team,

AI investigation detected a settlement failure for trade TRD1002c3.

Message Type: MT647

Root Cause: **Insufficient Securities**

Category: **SECURITY**

Severity: **HIGH**

Recommended Action: Acquire additional securities before settlement deadline.

Action orchestrator

Action Orchestrator

AI triggered 3 operational actions across enterprise systems.

JIRA COMPLETED

P2 Settlement Failure - TRD1002c3

JIRA-TRD1002c3-SECURITY

TEAMS COMPLETED

Settlement Alert - TRD1002c3

TEAMS-TRD1002c3

EMAIL COMPLETED

Escalation Email - TRD1002c3

EMAIL-TRD1002c3

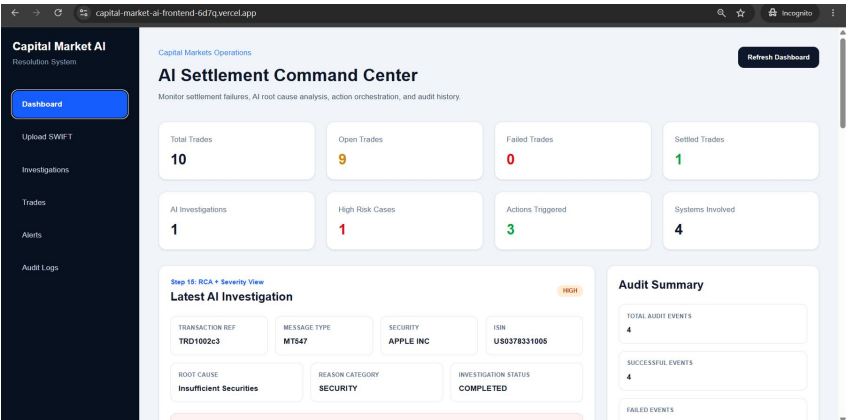
Generated Outcome

- Jira ticket created
- Teams alert generated
- Email escalation previewed
- Audit trail updated

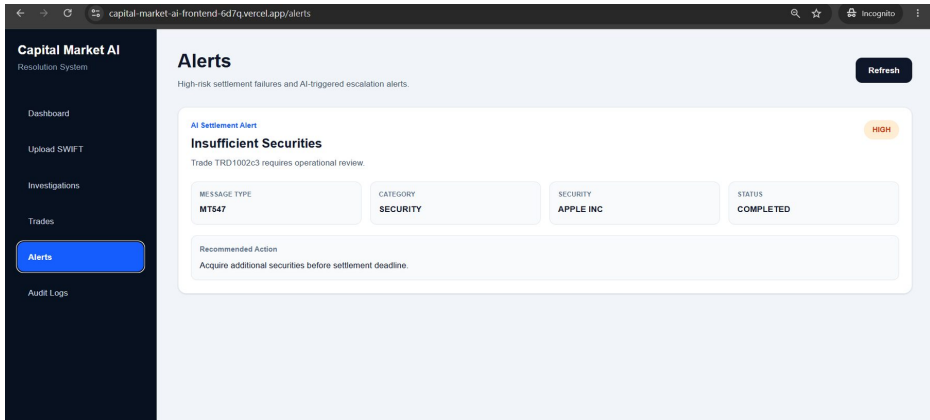
Demo Screenshots: Dashboard, Alerts & Audit

Operational visibility for investigations, risk, alerts, and traceability.

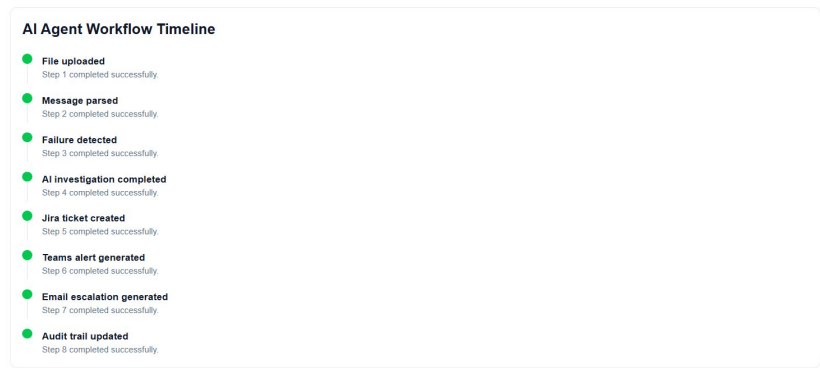
Command center



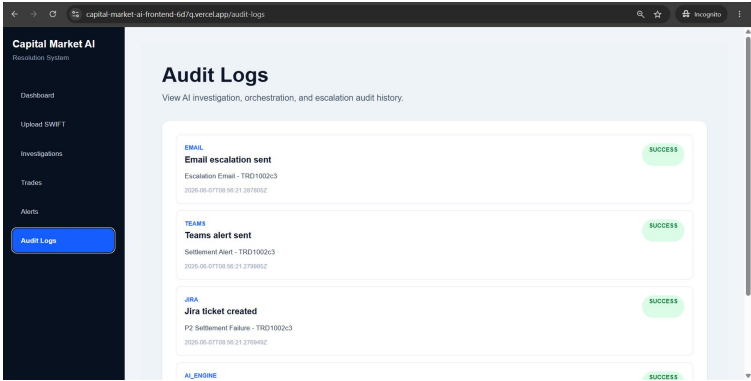
High risk alert



Agent workflow timeline



Audit logs



Impact, Tech Stack & Team

Built as a working prototype for the hackathon submission.

Impact

- Reduces manual investigation effort
- Faster root cause analysis
- Consistent recommended actions
- Traceable audit history

Microsoft AI Stack

- Azure OpenAI
- Semantic Kernel
- Azure App Service
- Azure Blob / SQL-ready design
- Azure AI Foundry-ready

Team

Anubhav Bangari

Full Stack Engineer

Capital Markets + AI Builder

Role: End-to-end product, backend, AI, UI

Submission Checklist Covered

● Problem statement

Slide 2

● Solution overview

Slide 3

● Architecture diagram

Slide 4

● AI integration details

Slide 5

● Demo screenshots

Slides 6-9

● Team introduction

Slides 1 and 10